



Counterfactual ensembles for interpretable churn prediction: from real-world to privacy-preserving synthetic data

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Abstract

Counterfactual explanations identify minimal input changes needed to alter a machine learning model's prediction, offering actionable insights in tasks like churn analysis. However, existing methods often produce counterfactuals that vary in quality, coherence, and plausibility, limiting their practical value. We propose an ensemble evaluation framework that integrates multiple generation techniques and ranks their outputs using a tunable scoring function balancing multiple relevant metrics. Our approach addresses two key deployment scenarios: (i) in-house churn analysis, where decision-makers can interactively adjust scoring weights for tailored, user-driven explanations; and (ii) outsourced churn prediction, where counterfactuals must be generated on synthetic data to preserve privacy while remaining representative of real cases. Experiments on benchmark churn datasets demonstrate that our ensemble approach improves the consistency, interpretability, and utility of counterfactuals across both real and synthetic settings, supporting more reliable and privacy-aware decision-making.

Keywords Explainable AI · Counterfactual explanations · Churn analysis

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1 Introduction

Customer churn, the decision by clients to terminate a service, is a pervasive and costly issue in many industries, particularly those offering subscription-based or contract-driven services. Identifying which customers are likely to churn is only the first step; businesses must also understand the underlying reasons behind customer attrition and how to intervene effectively. This makes churn analysis not just a predictive task, but a high-stakes decision-making problem where interpretability and actionable insights are essential.

In this context, Explainable Artificial Intelligence (XAI) has emerged as a powerful approach to bridge the gap between model accuracy and human understanding. While machine learning (ML) models have demonstrated high performance in predicting churn, their increasing complexity often comes at the cost of transparency. As a result, many models become opaque to the practitioners who rely on them, creating significant barriers to trust, accountability, and regulatory compliance (Ali et al., 2023). These concerns are especially pressing in domains like churn analysis, where decisions directly affects customer engagement strategies and, ultimately, revenue.

XAI methods, and counterfactual explanations in particular, offer an intuitive and legally relevant¹ solution (Bodria et al., 2023). By identifying the minimal changes in input features required to flip a model's prediction, counterfactual explanations support contrastive reasoning—answering the question: “What would need to change in a customer's profile to prevent churn?” This form of explanation is not only intuitive but aligns with emerging legal requirements, such as those established under the GDPR² and the AI Act³, concerning the right to explanation and transparency (Stepin et al., 2021; Guidotti, 2022). In the context of churn analysis, counterfactuals provide actionable recommendations that help retention officers develop personalized intervention strategies (Geiler et al., 2022; Joy et al., 2024).

However, the quality of counterfactuals can vary significantly across different generation techniques, with each method optimizing for different properties such as sparsity, plausibility, or diversity (Nauta et al., 2023). This heterogeneity poses a challenge when integrating XAI into practical churn workflows, especially in the absence of a standardized evaluation methodology tailored to the domain.

To address this challenge, we propose an ensemble evaluation framework specifically designed for churn prediction. Our framework integrates multiple counterfactual generation methods and ranks the resulting explanations using a dynamic, customizable scoring function. This function balances key counterfactual properties—proximity, sparsity, plausibility, and diversity—to surface the most relevant and realistic explanations with minimal computational overhead (Guidotti & Ruggieri, 2021). This design empowers practitioners to tailor explanations to the specific demands of churn analysis, where certain properties such as feasibility and minimality are particularly crucial.

Our approach supports two primary operational scenarios. In the first, churn analysis is conducted in-house, and the interpretability interface enables user-driven exploration. By tuning the scoring function, decision makers can prioritize explanation characteristics that align with business goals, facilitating targeted, customer-specific interventions based on

¹For Reference see GDPR articles 12-15 and 22.

²See previous footnote.

³For Reference see AI Act articles 13, 50 regarding transparency provisions for high-risk system providers and deployers and Article 86 regarding the right to explanation for individual decision-making.

model output. In the second scenario, the prediction of churn is outsourced to third-party providers⁴. These services can include machine learning-based churn prediction, analysis of explanations, and intervention strategy design. In such cases, privacy considerations prevent the direct sharing of real customer data. To overcome this constraint, we extend our framework to operate in privacy-preserving settings by employing synthetic datasets that retain the statistical properties of real data while eliminating personally identifiable information (Qian et al., 2024). While generally faithful to the real counterpart, synthetic data do not guarantee faithful counterfactuals. To ensure that synthetic counterfactuals remain aligned with their real-data counterparts, we introduce an optimization strategy for our ensemble method that minimizes the Kullback-Leibler (KL) divergence between the distributions of selected counterfactuals from real and synthetic sources. This alignment ensures that third parties receive explanations that are both representative of real situations and privacy-compliant.

We validate our ensemble framework using four publicly available churn datasets, demonstrating that it improves the robustness, interpretability, and practical utility of counterfactual explanations. Our results show that the method is effective in both in-house and privacy-sensitive scenarios, supporting trustworthy, human-centered decision making in customer retention workflows.

1.1 Contributions

This work makes the following key contributions:

- **Domain-specific ensemble evaluation framework:** We introduce an ensemble-based framework for evaluating and selecting counterfactual explanations tailored to churn prediction, integrating multiple generation methods and ranking them using a scoring function.
- **Customizable multi-criteria selection:** The scoring function balances proximity, sparsity, plausibility, and diversity, enabling practitioners to adapt the evaluation process to domain-specific needs.
- **Support for user-driven counterfactual exploration:** In the in-house setting, our framework allows interactive control over explanation generation, supporting interpretable, client-specific interventions.
- **Privacy-preserving explanation generation with distributional alignment:** For the outsourced prediction of churn, we enable the use of synthetic data to protect personal information while ensuring the fidelity of the explanation. This is achieved through a KL-divergence-based optimization strategy that aligns the distributions of real and synthetic counterfactuals.
- **Comprehensive empirical validation:** We validate the approach in four public churn datasets, showing improved robustness, interpretability, and compliance with privacy in both internal and external deployment settings.

⁴Outsourcing Churn Analysis and Retention is a common practice. Companies specialized in Churn Analytics include InfoSysBPM, iQor and Konsyq, etc.

2 Related works

Early research on churn modeling has primarily emphasized predictive accuracy, employing a range of supervised and unsupervised learning methods to analyze customer behavior, transaction histories, and demographic variables (Burez & Poel, 2008; Maldonado et al., 2020; De Bock & De Caigny, 2021; Adhikary & Gupta, 2020). These models often perform well in identifying likely churners, but their complexity tends to obscure the underlying decision logic, limiting their usefulness in real-world retention strategies (Dong et al., 2018; Mishra & Reddy, 2017; Petkovski et al., 2016).

To address the interpretability gap, counterfactual explanations have emerged as a promising post-hoc approach within Explainable AI (XAI). These methods identify the minimal changes needed in a feature vector to alter a model's decision (Bodria et al., 2023). Although similar in structure to adversarial examples, counterfactuals are intended for human consumption and emphasize informative properties such as proximity, sparsity, plausibility, and diversity (Freiesleben, 2022; Guidotti, 2022).

Counterfactual reasoning is especially well-suited to personalized decision-making contexts, including churn management, as it allows businesses to identify individual-level interventions that may prevent customer loss (Lemon & Verhoef, 2016; Tung, 2024). However, most XAI applications in churn analysis have relied on feature importance techniques, such as SHAP or LIME (Joy et al., 2024; Theodoridis & Tsadiras, 2022; Xiong et al., 2023), with relatively few studies exploring the use of counterfactuals for this domain.

From a methodological standpoint, optimization-based approaches to counterfactual generation are well-established (Carrizosa et al., 2024; Kanamori et al., 2021; Tan et al., 2021). However, existing studies typically focus on single-instance generation and do not consider ensemble-based selection strategies. To the best of our knowledge, no prior work has systematically applied ensemble evaluation of counterfactuals to churn prediction.

Privacy for churn prediction and analysis has been a focus of several works. In the article by Coimbra et al. (2024), edge computing is used to process data for Churn analysis locally, without transmitting users data. A similar approach is adopted by Huh and Lee (2024), where privacy is guaranteed via Federated Learning, specifically in the telecommunication sector. In both these works, the focus is in avoiding data transmission between nodes, thus preventing sensible information from being shared outside of the first-party managing the churn prediction. This suggests, as it happens in many practical cases, that churn prediction is either performed directly by the data owner or, in case of outsourcing, is performed using synthetic data, generated to replicate the statistical properties of real datasets while obfuscating personal identifiers (Assefa et al., 2020). Empirical evidence from domains such as healthcare suggests that models trained on synthetic data can yield comparable performance to those trained on real-world data (Rankin et al., 2020). Moreover, synthetic data generation is increasingly being combined with differential privacy guarantees to enhance protection without compromising utility (Bellovin et al., 2019). Hyrup et al. (2025) provides a comprehensive overview of privacy-preserving techniques for synthetic data in the context of health related data. Although a different domain, it still provides a good base to understand how synthetic data can be used to enhance privacy in any data analysis or machine learning process. The protection offered by synthetic data can still however be foiled, as proved by Ganey and De Cristofaro (2025). In this work, the authors devise an attack model that leverages the weaknesses of distance based privacy metrics to foil the protection of

synthetic data. However, the adversarial model adopted in the paper assumes a worst-case scenario analysis, thus requiring the leaking of a lot of different data and parameters in order to be effective.

3 Problem definition

3.1 Background

Churn prevention encompasses all the actions that a company puts into place to prevent the loss of customers. The first and most important part of any churn prevention strategy lies in detecting which customers will likely interrupt their relationship with the company, given their current status. A churn officer has then the duty of interacting with these customers in order to find possible actions to prevent them from churning. This task can be modeled as a binary classification task (Geiler et al., 2022) where a machine learning model is used to predict which customers are the ones likely to churn.

Formally, given a feature space $\mathcal{X} \subseteq \mathbb{R}^d$ where each customer is represented by a d dimensional vector $x = (x_1, \dots, x_n) \in \mathcal{X}$ and a label space $\mathcal{Y} = \{0, 1\}$ where 1 indicates a customer churning and 0 a customer not churning, we assume to observe a dataset of N i.i.d samples $\mathcal{D} = \{(x_i, y_i)\}_{i=0}^N \stackrel{\text{iid}}{\sim} P(x, y)$ over $\mathcal{X} \times \mathcal{Y}$. A churn predictor (or classifier) is a function $f : \mathcal{X} \rightarrow \mathcal{Y}$ usually trained over a training set $\mathcal{D}_{train}$ via empirical risk minimization.

In this context, counterfactual explanations are extremely useful from a business' perspective: firstly, counterfactual explanations help identify the minimal changes x needed to retain a customer; for example, if the model indicates that a customer is likely to churn, a counterfactual explanation might reveal that offering a small discount, offering a particular product or improving service quality could change the prediction to retention. This allows businesses to implement precise interventions that are cost-effective and efficient. Secondly, the interpretability offered by counterfactual explanations builds trust among business stakeholders. Unlike explanatory methods that might provide abstract or general insights, such as in the case of global explanations, counterfactuals show specific scenarios and outcomes, making it easier for non-technical stakeholders to understand and trust the model's recommendations. This trust is vital in securing support and commitment from stakeholders for data-driven strategies and ensuring their successful implementation. Counterfactual explanations have not been explored for the problem of specific optimization strategies in churn prevention (Joy et al., 2024; Theodoridis & Tsadiras, 2022; Xiong et al., 2023).

Formally, given an instance x and the corresponding churn prediction $f(x) = y$ a counterfactual explanation for x is a point x' such that $f(x') = y'$ with $y \neq y'$, i.e. the classifier's output flips from its original prediction from y to y' . To find a minimal counterfactual, given a distance $d(x, x^*)$, we can solve the optimization problem:

$$\begin{aligned} x' = & \arg \min_{x^*} d(x, x^*) \\ \text{s.t. } & f(x) \neq f(x') \end{aligned} \quad (1)$$

where $d(x, x^*)$ is commonly chosen to be the (L_1) or (L_2) -norm. Many counterfactual explanation methods. However, depending on what counterfactual method one chooses (Guidotti, 2022), the explanations obtained may rely on specific optimization strategies that are variations or approximations of formulation 1 and therefore explore only some particular aspect of the importance of a churn officer. Moreover, many counterfactual methods produce multiple counterfactual instances for any given x . We can therefore define $X' = \{x'_1, \dots, x'_v\}$ as the set of valid counterfactuals produced by any method for instance x .

3.2 k-CEM: counterfactual ensemble method

To tackle similar problems, Guidotti et al. (Guidotti & Ruggieri, 2021) proposed an ensemble method that leverages the strengths of multiple counterfactual explainers to cover a set of desirable properties, such as minimality, actionability, stability, diversity, plausibility, and discriminative power. Their approach demonstrates the efficacy of boosting weak explainers into a powerful ensemble that is both model-agnostic and data-agnostic, capable of handling various data types including tabular data, images, and time series.

Building upon this idea, we propose an ensemble approach that operates ex-post as an evaluation and selection mechanism, called *k-CEM*. Our method is designed to identify the optimal set of counterfactual examples by employing a linear combination score of various metrics, that reflect on the possible aspects that a churn officer would explore in a churn prediction model. In contrast to the ensemble proposed by Guidotti and Ruggieri (2021), which combines results through a diversity-driven selection function, our framework introduces a more nuanced selection score. This approach not only refines the selection process but also ensures that the chosen counterfactuals align closely with the desired properties - thereby improving the interpretability and reliability of the explanations provided – and that can be aptly tweaked by practitioners to give more emphasis to a specific metric. Churn analysis, however, is often outsourced to specialized organizations. In such circumstances, privacy concerns often demand the use of protection mechanism, such as for example the use of synthetic data. Synthetic data can be shared outside the organization for analysis (Hyrup et al., 2025). However, while synthetic data is generated from real data and therefore similar in general properties, it usually provides less accurate models and far less reliable explanations. This means that third-party churn analysts cannot devise realistic customer retention strategies based on these explanations. Thanks to the selection function that we devise, it is possible to use k-CEM to improve churn prediction procedure in such a privacy-aware setting by providing a tool to interact with outsource churn analysts without compromising the quality of the explanations. To do this, we can exploit the selection function on synthetic data with a KL-minimization strategy (Section 3.4) to find the best parameters to produce explanations that are as close as possible to realistic explanations on real data. Let us define set of counterfactual explanation methods as $E = \{e_1, \dots, e_q\}$ where $e_i : (f, x, y') \rightarrow x'$. With slight abuse of notation, we indicate with $E(f, x, y') = \{e_1(f, x, y'), \dots, e_q(f, x, y')\}$. Our proposal is to score the counterfactual explanations produced by an ensemble of counterfactual methods E using evaluation metrics that align with desired properties in the context of Churn analysis. The pseudocode of our approach is given in Algorithm 1.

```

1: Input:  $x$  - instance to explain,  $E$  - set of CF explanation methods, size of top
   ensemble  $k$ , set of metrics  $M$ , churn prediction model  $f$ , target label  $y'$ .
2: Output:  $C$  - ensemble of top counterfactual explanations
3:  $C \leftarrow \emptyset$ 
4:  $X' \leftarrow E(f, x, y')$ 
5: Users chooses  $score \leftarrow linear \mid hierarchical$ 
6: while  $user_{stop} = False$  do
7:   for  $x' \in X'$  do
8:     Calculate and normalize metrics:  $m_i(x', X')$  with  $0 < i < |M|$ 
9:     User chooses  $w_i$  with  $0 < i < |M|$ 
10:    if  $score = linear$  then
11:       $score(x') \leftarrow \sum_i^{|M|} w_i m_i$ 
12:    else if  $score = hierarchical$  then
13:       $score(x') \leftarrow \bigcup_{i=1}^{|M|} \mathcal{S}_i$ 
14:    end if
15:  end for
16:   $Sort(X', score, ascending)$ 
17:  Select top- $k$   $x' \in X$ 
18:  if user is satisfied then
19:     $user_{stop} \leftarrow True$ 
20:  end if
21: end while
22: Return  $C$ 

```

Algorithm 1 k-CEM: k-Counterfactual Ensemble Method

For any given instance, once a set of valid counterfactual is obtained, k-CEM allows the user to define a selection score (Section 3.3) and weights of a set of metrics M . The priority of each metric is user-defined, therefore the user is able to modify the scores and interact with the explanations, without having to recompute them. This point is crucial, and is the reason for the *ensemble* of methods E : by increasing the number and diversity of counterfactuals, the selection phase of k-CEM is able to exploit the representativeness of the counterfactual set X' without overloading the user with information, presenting a set of explanations that answer specific user needs as defined by the metrics M . The set of parameters w_i is main "handle" that a user can leverage to interact with k-CEM, modifying priorities of metrics and exploring the space of explanations. We remark that all the explanations in k-CEM are valid, local explanations. While global or group-level counterfactuals (Warren et al., 2024) offer scalability and coherence, our method deliberately adheres to instance-specific counterfactuals to guarantee perfect validity and actionability in a customer-retention context. Group explanations risk diluting personalized recommendations—potentially yielding suggestions that are suboptimal or invalid for individuals at the periphery of a heterogeneous subgroup. With our method, we ensure semantic consistency and predictive fidelity, while remaining compatible with privacy-preserving techniques (e.g., synthetic data) to safeguard confidentiality without compromising the precision of each explanation. The framework in which k-CEM is applied is described in Fig 1. We envision k-CEM to be at the center of two possible usage settings: *first-party* churn analysis or *third-party* churn analysis.

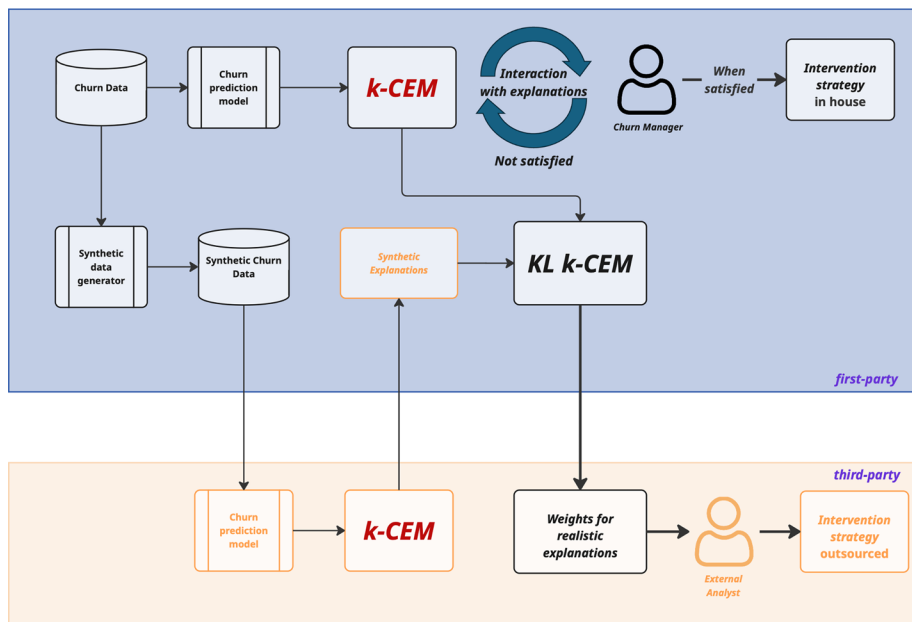


Fig. 1 Churn analysis and explanation workflow with k-CEM

- **First-party** In first-party mode, k-CEM can be used by an organization internal personnel to perform in-house churn analysis, based on machine learning models trained directly on the data gathered and managed by the organization. In this scenario, the selection score of k-CEM allows the internal churn manager to interact with the explanations, modifying the selection score (Section 3.3) to obtain the explanation that best suit the needs of the organization. This allows the churn manager to develop personalized intervention strategies by interacting with k-CEM at a reduced computational cost.
- **Third-party** In third-party mode, k-CEM can be used to interact with an external analyst that can devise an intervention strategy based on shared synthetic data. The organization needs only to share the synthetic dataset and the specifics of k-CEM, i.e., the set of metrics of interest, to the third-party analyst. The analyst will then find the best machine learning model to predict churn on the synthetic data and use k-CEM to explain such a model. The organization can then use the model and explanations provided by the external analyst to adjust k-CEM via KL-minimization strategy (Section 3.4) to find a set of parameters that can provide realistic explanations for the synthetic data. Therefore the external analyst can build an intervention strategy based on such explanations, without ever accessing real data or real explanations.

In the following we are going to define the selection scores (Section 3.3) and KL-minimization strategy (Section 3.4) needed to make k-CEM function in the framework we devised, and we will provide the relevant counterfactual methods E and metrics M relevant to churn prediction (Section 3.6)

3.3 Selection scores

In this work, we propose two distinct strategies for combining evaluation metrics: a *linear selection score* and a *hierarchical selection score*, as detailed below. The specific metrics and counterfactual generators underlying these scores are introduced in later sections, where they are defined in accordance with the application context.

3.3.1 Linear selection score

The selection score of each counterfactual explanation can be computed as a weighted linear combination of chosen metrics m_i , which are defined so that a lower value indicates a higher counterfactual performance. This formulation allows a flexible synthesis of the most relevant aspects of counterfactual explanations based on application-specific priorities.

Formally, the score is defined as:

$$\text{Linear Score} = \sum_i^{|M|} w_i m_i \quad (2)$$

where w_i are user-defined non-negative weights summing to 1, m_i represents the i -th chosen counterfactual metric and $|M|$ is the number of the considered metrics. The final set of counterfactual explanations is obtained by ranking candidates according to their selection score in ascending order and selecting the top k instances, thus prioritizing explanations that best align with the weighted optimization criteria.

3.3.2 Hierarchical selection score

In addition to the linear combination, we propose a hierarchical selection score designed to prioritize metrics sequentially based on their relative importance. Rather than computing a single composite score by summing weighted metric values, our hierarchical procedure operates in two distinct phases: allocation and refinement. Let M denote the number of evaluation metrics, $\mathcal{D}$ the set of candidate counterfactuals, and k the total number of explanations to select. We assume normalized weights w_i for each metric m_i , such that $\sum_{i=1}^{|M|} w_i = 1$. For each metric, we then apply a ranking-based selection:

$$\mathcal{S}_i = \text{Top}(k_i, \mathcal{D}, \text{metric}_i) \quad (3)$$

where $\text{Top}(k_i, \mathcal{D}, m_i)$ returns the k_i candidates from $\mathcal{D}$ that optimize metric_i , and $k_i = \lfloor k \cdot w_i \rfloor$. We aggregate these sets:

$$\mathcal{S} = \bigcup_{i=1}^{|M|} \mathcal{S}_i, \quad (4)$$

Because each metric draws from the same pool $\mathcal{D}$, overlaps can occur: $|\mathcal{S}|$ may exceed k . To meet the overall budget, we apply a refinement step. We order the aggregated set according

to the primary (highest-weight) metric, breaking ties by the next metric in weight order, and truncate to the top- k instances. This ensures that any over-selection is resolved by prioritizing the most important metric, then the next, and so on.

3.3.3 Comparison of scores

The Linear selection scores pools all metrics at once into a continuous score $s(x) = \sum_i w_i m_i(x)$ and selects the top- k by that score. In contrast, our hierarchical approach discretizes the weight budget into per-metric quotas and enforces a minimal representation from each metric before considering composite performance. If a single weight $w_j > 0.5$, the hierarchical method reserves $\lfloor 0.5k \rfloor$ slots exclusively based on metric j , guaranteeing that at least half the explanations excel in that metric. In a pure linear score, setting $w_j = 0.5$ similarly emphasizes j , but does not enforce a minimum count—other metrics could crowd out metric j if their combined scores exceed those of metric- j -focused instances. The hierarchical allocation and refinement scheme thus provides interpretable, controllable guarantees on per-metric representation, complementing the more fluid trade-offs of linear combination.

3.4 KL divergence minimization for parameter tuning

In privacy-preserving settings, particularly when churn analysis is outsourced and real data cannot be directly used, it's essential that synthetic counterfactual explanations closely mirror those derived from real data. To address this challenge, we introduce a parameter tuning procedure based on minimizing the Kullback–Leibler (KL) divergence between the feature distributions of the real and synthetic counterfactual ensembles. Formally, we measure the distance between the feature distributions on real ($\mathcal{R}$) and synthetic ($\mathcal{S}$) datasets using the average Kullback–Leibler (KL) divergence:

$$\bar{D}_{\text{KL}}(\mathcal{S} \parallel \mathcal{R}) = \frac{1}{n} \sum_{i=1}^n D_{\text{KL}}(Q_i \parallel P_i) \quad (5)$$

where n is the number of features considered, and Q_i, P_i denote the empirical distributions of the i -th feature in the synthetic and real counterfactual datasets, respectively. For continuous variables, P_i and Q_i are estimated via histogram binning, with the number of bins determined by Sturges' rule to balance bias and variance. In this minimization, we jointly optimize the weight vectors $w_{\mathcal{R}} = \{w_1^{\mathcal{R}}, w_2^{\mathcal{R}}, w_3^{\mathcal{R}}, w_4^{\mathcal{R}}\}$ and $w_{\mathcal{S}} = \{w_1^{\mathcal{S}}, w_2^{\mathcal{S}}, w_3^{\mathcal{S}}, w_4^{\mathcal{S}}\}$, which govern the selection score functions for the real and synthetic datasets, respectively. Alternatively, one may fix $w_{\mathcal{R}}$ according to user-defined preferences and optimize only $w_{\mathcal{S}}$, thereby preserving user-driven interpretability in the real counterfactuals while allowing for privacy-preserving adaptation in the synthetic ones. However, to avoid overcomplicating the narrative, we focus in this work on the joint optimization of both $w_{\mathcal{R}}$ and $w_{\mathcal{S}}$. The optimization problem is formulated as:

$$\min_{w_{\mathcal{R}}, w_{\mathcal{S}}} \bar{D}_{\text{KL}}(\mathcal{S}(w_{\mathcal{S}}) \parallel \mathcal{R}(w_{\mathcal{R}})), \quad (6)$$

where $\mathcal{S}(w_{\mathcal{S}})$ and $\mathcal{R}(w_{\mathcal{R}})$ denote the distributions of features of the counterfactuals selected by the respective weight configurations. We solve this problem using a constrained optimization framework based on trust-region methods, which balance local model fidelity and global convergence guarantees. The optimization is performed under the normalization constraint $\sum_i w_i = 1$ for each weight vector, ensuring interpretability and comparability of the resulting scores. To enhance robustness and mitigate sensitivity to local minima inherent in the non-convex landscape of $\bar{D}_{\text{KL}}$, the optimization is repeated over 100 independent random initializations. In each trial, the trust-region algorithm iteratively adjusts the weights within adaptive neighborhoods, progressively refining the solution toward a local minimum. The final parameter configuration is selected as the one yielding the minimal KL divergence across all runs. This procedure provides a principled approach to calibrate the evaluation metrics, aligning synthetic data-driven counterfactual explanations with their real-data counterparts.

3.5 Coherence assessment via cluster-based distribution alignment

While the KL minimization procedure can be applied to the whole population of counterfactuals generated (in both real and synthetic data), counterfactuals in churn analysis are often used to interpret individual behavior and devise client-retention strategies.

In light of this, we build upon the KL Divergence minimization framework introduced in Section 3.4, and we develop a cluster-based evaluation framework to assess the coherence of counterfactual explanations derived from real and synthetic datasets. The procedure begins by partitioning both real and synthetic datasets into distinct subgroups based on observable behavioral patterns (e.g., the actual churn outcome). This stratification ensures that subsequent analyses are performed on contextually analogous subsets, thereby enabling fair comparisons between the two data modalities. Within each behavioral subgroup, we apply the K-Means clustering algorithm to independently group the real and synthetic individuals into five clusters. These clusters serve as the basis for forming initial baskets of counterfactual explanations. Counterfactual explanations are computed for each cluster (real and synthetic) and KL Divergence minimization is performed to identify, for each pair of synthetic and real counterfactual baskets, the counterfactual explanations that are most similar in terms of their feature distributions. Subsequently, we quantify the coherence between the real and synthetic counterfactuals by comparing the clusters of data points and their respective counterfactual baskets. Two key measurements are obtained: the average Euclidean distance among the selected couples of synthetic and real counterfactual explanations (counterfactual-basket distance), and the Euclidean distance between the centroids of the corresponding synthetic and real clusters of individuals (centroid distance). By jointly analyzing these distances, we derive a nuanced metric of alignment wherein lower counterfactual-basket distances, combined with smaller centroid separations, indicate a higher degree of coherence. A schematization of this procedure can be found in Fig. 2.

3.6 Relevant generator and metrics for churn analysis

For the implementation of our Counterfactual Ensemble Method, we choose four different counterfactual generation methods that, in our opinion, condense the most diverse approaches to the generation of synthetic counterfactual explanations.

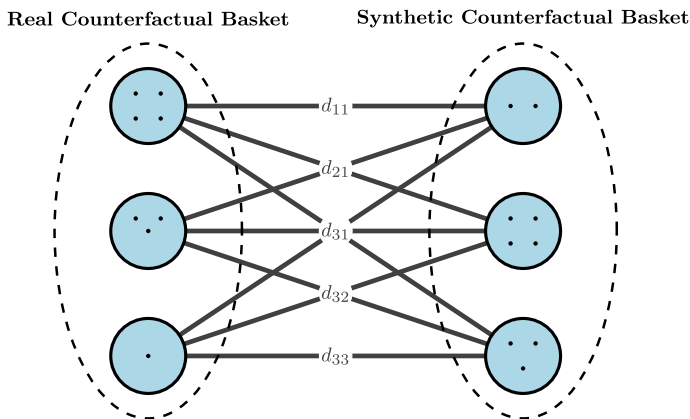


Fig. 2 Bipartite network illustrating potential matches between baskets of counterfactuals from real and synthetic datasets. Each node aggregates counterfactual explanations derived from clustered individuals, and the matching is assessed using the average Euclidean distance d_{ij} across all counterfactual components

- **DiCE** perturbs input features within the decision boundaries of the model, utilizing a genetic algorithm to create multiple instances that lead to different predictions (Sharma et al., 2020). It generates diverse counterfactual examples solving an optimization problem that balances properties of proximity and diversity.
- **Growing Spheres (GS)** uses a sphere-growing algorithm to iteratively explore the feature space around a given instance (Laugel et al., 2019). In our approach, we slightly modify GS to return the best k instances instead of just one, ranking them based on L_2 proximity to the original instance.
- **CFRL** is a model-agnostic counterfactual generation method that uses reinforcement learning (Samoilescu et al., 2021) to train a generative model to produce counterfactual explanations.
- **T-LACE** is a counterfactual explanation method that constructs a transparent latent space using a linear transformation where also the original prediction of the model is added, ensuring that similar records in the latent space have similar features and predictions (Bodria et al., 2022). Counterfactuals are then searched in the latent space decomposing contributions from each feature to identify a prediction direction.

The scoring function of the counterfactual ensemble constitutes the central component of our framework. It is grounded in properties that correspond to the practical considerations a churn analyst must address to mitigate the risk of customer attrition. When applied to alternative tabular data domains, such as credit risk assessment, this approach necessitates the selection of domain-specific metrics and their appropriate contextualization. In the following, we detail the metrics adopted in the churn prediction setting and elucidate their intended roles.

3.6.1 Proximity measures

How minimal are the changes required to retain potentially churning customers? Proximity measures indicate close counterfactuals. Proximity (also known as minimality (Byrne, 2019)) is a fundamental property of counterfactual explanations. We choose an average proximity measure using a geometric mean that combines various normalized proximity measures. The geometric mean prevents skewing by outliers, ensuring equal contribution from all proximity measures. The individual proximity metrics we use are:

Euclidean Distance (L_2 norm) measures the overall difference in feature values:

$$\text{Proximity}_{L_2} = \sqrt{\frac{m-h}{m} \sum_{i \in \text{cont}} (x'_i - x_i)^2 + \frac{h}{m} \sum_{j \in \text{cat}} \delta(x'_j, x_j)}$$

Manhattan Distance (L_1 norm) measures the sum of absolute differences:

$$\text{Proximity}_{L_1} = \frac{m-h}{m} \sum_{i \in \text{cont}} |x'_i - x_i| + \frac{h}{m} \sum_{j \in \text{cat}} \delta(x'_j, x_j)$$

Maximum Absolute Difference L_∞ norm measures the maximum element-wise absolute difference:

$$\text{Proximity}_{L_\infty} = \max \left(\frac{m-h}{m} \max_{i \in \text{cont}} |x'_i - x_i|, \frac{h}{m} \max_{j \in \text{cat}} \delta(x'_j, x_j) \right)$$

Here, m is the total number of features, h the number of categorical features, cont continuous features, cat categorical features, and $\delta(x'_j, x_j)$ is 1 if $x'_j \neq x_j$ and 0 otherwise (Hamming distance).

3.6.2 Plausibility measure

Is the counterfactual explanation similar to a non-churning customer in the data and thus justifiable to the customer? Plausibility indicates counterfactuals that have close examples in the original dataset.

The plausibility measure (also known as feasibility (Artelt et al., 2021)) assesses the degree of plausibility or soundness of the counterfactual instances (X') with respect to the instances in the original dataset to explain. Specifically, it calculates the minimum distance of each $x' \in X'$ from its closest instance in the original data. To compute the plausibility measure we build a KDTree (Maneewongvatana & Mount, 1999) on the X_{test} dataset to efficiently find the nearest neighbors, then we query the KDTree to find the nearest neighbor in X_{test} for each instance in the set of counterfactual instances and we calculate the distance between each x' and its closest instance in X_{test} . The use of a KDTree for computing the plausibility measure is motivated by the need to efficiently find the nearest neighbors of counterfactual instances within the dataset X_{test} . KDTree provides logarithmic search time complexity for nearest neighbor queries, making it more scalable compared to linear search methods, which have linear time complexity. Efficiency is crucial when the number

of instances in X_{test} is large, a common situation for real-world applications like churn analysis. Plausibility is represented as the euclidean distance of x' from its closest instance in the X_{test} population. A comparison between KDTree and the brute-force distance computation method is provided in the supplementary material, demonstrating the efficiency benefits of the KDTree approach, maintaining high output accuracy.

3.6.3 Sparsity measure

Does the counterfactual explanation modify as few features as possible, thus making the required changes easier for the churn officer to propose? Sparsity indicates counterfactuals that touch the least amount of features.

Sparsity (Guidotti, 2022) is computed as the fraction of differing features to the total number of features n :

$$\text{Sparsity} = \frac{\sum_{i=1}^n (x'_i \neq x_i)}{n} \quad (7)$$

3.6.4 Diversity measure

The counterfactuals produced do provide different courses of action for the churn officer? Diversity indicates that the explanations produced have enough variety for the churn officer to act on.

The diversity measure (Mothilal et al., 2020) quantifies the dissimilarity or variation within groups defined by the generation source. It is calculated as the mean of distances between pairs of instances within each group.

$$\text{Diversity} = \frac{1}{N} \sum_{i=1}^N \frac{1}{n_i(n_i - 1)} \sum_{j \neq k} d(x_j, x_k) \quad (8)$$

where N is the total number of groups - i.e. sets of counterfactuals for a given instance to explain -, n_i is the number of instances in group i , and $d(x_j, x_k)$ is the distance between instance j and instance k within the same group.

4 Experiments

For our experiments, we used four public datasets specifically focused on the churn classification problem. The "Credit Card Bank Churn", dataset⁵ includes 10000 credit card user records with 18 features to predict if a customer will stop using the bank's credit card services (0.19 ratio of churners). The "E-commerce Dataset"⁶ contains 5,630 customer records with 20 features, collected from a leading online retailer. It is used to predict customer churn, enabling targeted retention efforts through promotional offers (0.20 ratio of churn-

⁵<https://www.kaggle.com/datasets/anwarsan/credit-card-bank-churn>

⁶<https://www.kaggle.com/datasets/ankitverma2010/ecommerce-customer-churn-analysis-and-prediction>

ers). The "Iranian Churn Dataset"⁷ contains 3,150 records and provides telecommunications customer data from Iran with 13 features used to analyze churn behavior (0.18 ratio of churners) in the telecom industry. The "Telco Customer Churn" dataset⁸ contains information on 7,043 customers of a telecom provider, including service usage patterns, billing information, and demographic attributes. The dataset features a binary churn label and a churn rate of approximately 0.27, making it suitable for evaluating counterfactual explanations in real-world customer retention scenarios.

We start by identifying which model to explain. We compare the performance of Light Gradient Boosting Machine (LightGBM) (Ke et al., 2017), XGBoost (Chen & Guestrin, 2016), Random Forest (Breiman, 2001), and Multilayer Perceptron (MLP). Our focus is on the explanation methodology, more so than on the task, we therefore chose models that are commonly used for the task and easily applied (Geiler et al., 2022). To fine-tune the models, we conducted a randomized grid search with 5-fold cross-validation, optimizing for the ROC AUC score and we accounted for class imbalance penalizing errors on the minority class proportionally during training.

In Table 1 the comparison is displayed across datasets for F1-Score and Matthews Correlation Coefficient (MCC) measures. The LightGBM outperforms or at least performs as well as the XGBoost in the datasets under analysis while the Random Forest and MLP slightly underperforms. These results lead us to establish the LightGBM as the model of interest for the following counterfactual explanations.

4.1 Interactivity of the counterfactual ensemble for real data

We analyze the behavior of the ensemble selection framework under different configurations of both the linear and hierarchical scoring functions. Our objective is to understand if counterfactuals produced with the ensemble are aligned with practitioner-defined priorities and if the ensemble enables users to meaningfully interact with explanations. Our analysis focuses on two key experimental axes: (i) variation in the weight configurations assigned to the evaluation metrics, and (ii) stratification based on prediction confidence thresholds. We want, in other words, to verify if our method is responsive with respect to the possible

Table 1 Model Performance on Churn Datasets. The number of instances is referred to the test set only. LightGBM tends to perform better, while XGBoost and Random Forest display similar performances, and MLP consistently underperforms

Churn Dataset	Test Instances	Metric	LGB	XGB	RF	MLP
Card Churn	2026	F1 Score	0.87	0.86	0.80	0.73
		MCC	0.79	0.78	0.73	0.58
E-com Churn	755	F1 Score	0.95	0.89	0.91	0.81
		MCC	0.90	0.78	0.86	0.61
Iranian Churn	630	F1 Score	0.90	0.88	0.88	0.79
		MCC	0.87	0.74	0.85	0.65
Telco Churn	1408	F1 Score	0.71	0.72	0.72	0.70
		MCC	0.61	0.63	0.63	0.59

⁷ <https://archive.ics.uci.edu/dataset/563/iranian+churn+dataset>

⁸ <https://community.ibm.com/community/user/businessanalytics/blogs/steven-macko/2019/07/11/telco-customer-churn-1113>

choices of a potential user, i.e., how interactive is the basket of counterfactuals produced by the ensemble.

We begin by examining the effect of different weight allocations in the score function. Specifically, we consider five configurations: one with equal weighting across all four metrics ($w_i = 0.25$), and four imbalanced settings in which a single metric receives a dominant weight of 0.5, while the remaining three metrics are equally assigned a weight of 0.1667. For each configuration, we compute the selection score of every counterfactual generated by the ensemble and rank them accordingly. The top $k = 5$ counterfactuals are then selected for each instance and algorithm.

Figure 3 displays the proportion of counterfactuals originating from each method—DiCE, T-LACE, GS, and CFRL—that are selected under different weight configurations for four benchmark churn datasets and for both the linear and hierarchical selection scores. The ensemble exhibits strong adaptability. T-LACE is consistently favored in both the Credit Card Churn and E-commerce Churn datasets, regardless of the emphasized metric. In contrast, DiCE emerges as the top-performing generator in the Iranian Churn dataset, while CFRL dominates in the Telco dataset. These results confirm that the quality of counterfactual explanations varies significantly across methods and datasets, reinforcing the need for an ensemble approach that can dynamically tailor its selections based on user-defined priorities. In contrast, the hierarchical evaluation function demonstrates a pronounced impact of user-specified parameter interactions. Across all datasets, no single counterfactual generator exhibits consistent dominance. Instead, the interplay between evaluation metrics yields dynamic selection patterns: for example, in the Credit Card Churn dataset, a more balanced selection is displayed. The hierarchical approach shows no stable trend, with the top-ranked counterfactuals shifting markedly as the weights change, illustrating that the hierarchical approach facilitates a more nuanced and adaptive assessment of counterfactual quality.

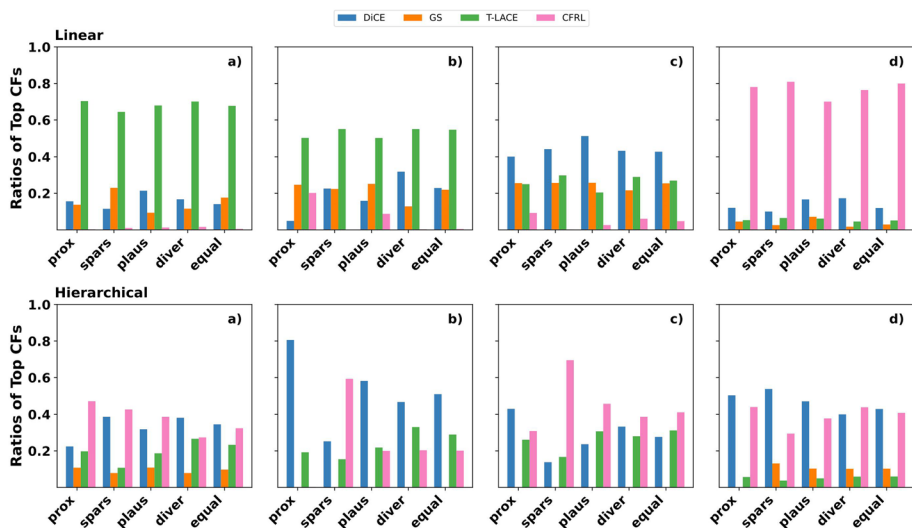


Fig. 3 Proportion of top-ranked counterfactuals selected by the ensemble across four datasets under different weight configurations: **a)** Credit Card Churn, **b)** E-commerce Churn, **c)** Iranian Churn, and **d)** Telco Churn. The x-axis indicates the metric with the highest weight. In the linear evaluation (top panel), DiCE dominates in **a)** and **c)**, T-LACE in **b)**, and CFRL in **d)**. The hierarchical evaluation (bottom panel) is highly sensitive to parameter changes, underscoring the impact of user-defined interactions

Moreover, we highlight that the hierarchical evaluation function is more intuitive in its formulation for a human, as it can be simply designed as a set of priorities.

To further investigate the relationship between counterfactual selection and data point characteristics, we stratify the candidate counterfactuals into three bins according to their predicted probability for the target class: $[0.5, 0.7)$, $[0.7, 0.9)$, and $[0.9, 1.0]$. This confidence interval reflects the certainty of the black-box model in assigning the counterfactual to the opposite class of the original instance. For each confidence bin and weight configuration, we rerank the counterfactuals using the linear score and retain the top five per instance. Figure 4 illustrates the proportion of selected counterfactuals for each generation method under this joint stratification.

Using the linear selection score, the figure reveals complementary insights. In the Credit Card dataset, T-LACE maintains a consistent advantage across all confidence levels, indicating its robustness to prediction certainty, whereas in the E-commerce dataset as the threshold increases, patterns vary significantly. In the Iranian and Telco Churn datasets, DiCE-generated counterfactuals not only dominate overall, but are increasingly selected as prediction confidence rises, suggesting that DiCE yields more reliable, high-confidence explanations in these contexts. In contrast, GS and CFRL are more frequently selected at

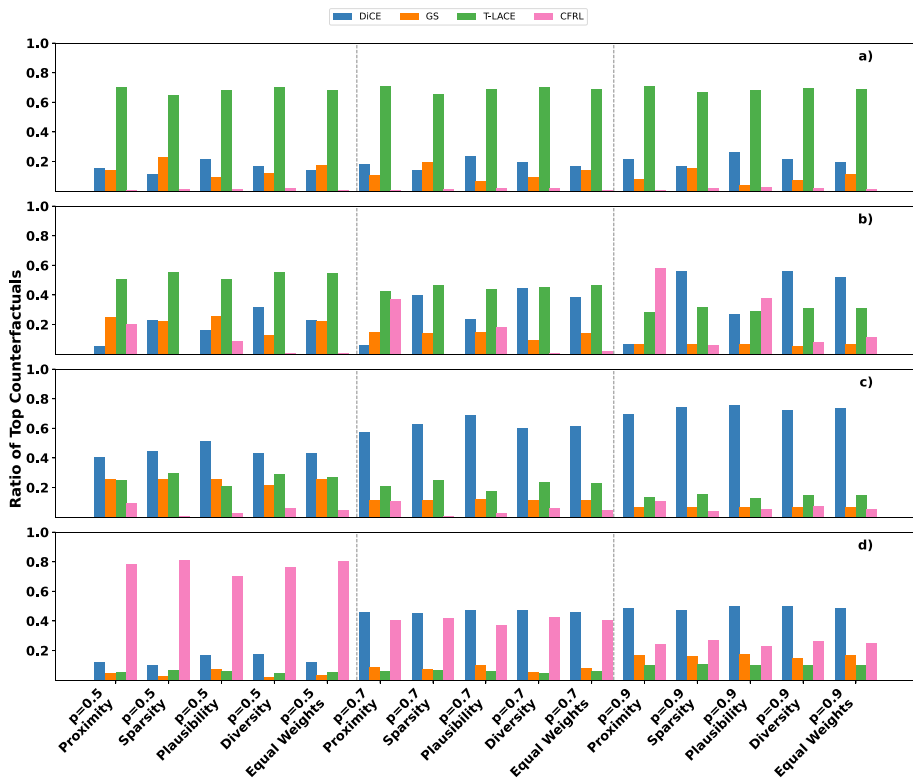


Fig. 4 Proportion of top-ranked counterfactuals stratified by prediction probability thresholds and weight configurations across datasets: **a)** Credit Card Churn, **b)** E-commerce Churn, **c)** Iranian Churn, and **d)** Telco Churn. DiCE remains dominant in **a)** and **c)** at higher confidence thresholds. T-LACE is consistently favored in **b)**. CFRL is selected more frequently in **d)** when confidence is low, but this preference diminishes at higher thresholds

lower confidence thresholds, highlighting their relevance to generate exploratory or less decisive alternatives. Interestingly, in the Telco Churn dataset, CFRL predominates only at lower confidence thresholds but is progressively supplanted by other methods as the threshold increases. The results for the hierarchical selection score are provided in the Supplementary Information Fig. 4 and are omitted here for length limits.

4.2 Comparison of synthetic and real counterfactual ensembles via KL divergence minimization

To minimize KL divergence between real and synthetic counterfactuals, we first selected a suitable synthetic data generator. We evaluated four state-of-the-art methods—CTGAN (Xu et al., 2019), GaussianCopula (Patki et al., 2016), TVAE (Xu et al., 2019), and CopulaGAN (Patki et al., 2016)—on our four churn datasets using default parameters. The data quality was assessed with the SDV library, which computes two indicators: Column Shapes (measuring how well each column’s marginal distribution is preserved via the Kolmogorov–Smirnov statistic for numerical/DateTime columns and total variation distance for boolean/categorical ones) and Column Pair Trends (evaluating the preservation of relationships between columns using correlation similarity for numerical/DateTime pairs and contingency similarity for boolean/categorical pairs, with numerical data binned when paired with categorical data). The overall quality score is the average of these two metrics, bounded between 0 and 1. As shown in Table 2, CTGAN achieved the highest average quality score (0.87) compared to GaussianCopula (0.83), TVAE (0.79), and CopulaGAN (0.83); hence, CTGAN was chosen for all subsequent experiments.

Once a proper synthetic data generator is chosen, we can further assess the fidelity of our ensemble counterfactual explanations by comparing those generated on synthetic data (via CTGAN) to those derived from real data. In privacy-preserving scenarios, it is essential that the synthetic ensemble reliably approximates the characteristics of the real-data ensemble. To quantify this alignment, we measure the average KL divergence, $\bar{D}_{KL}$, between the feature distributions computed on the entire population of real and synthetic counterfactuals. For each dataset, we construct ensembles of counterfactual explanations using the linear selection score (as defined in Sec. 3.3.1). To mitigate sensitivity to initialization and ensure the robustness of the solution, the optimization problem formulated in Section 3.4 is solved over 100 independent runs using trust-region methods. This procedure yields the optimal weight parameters for both the real (w_R) and synthetic (w_S) ensembles that minimize $\bar{D}_{KL}$. While the hierarchical function achieves increased levels of interactivity, the linear selection score guarantees lower level of KL Divergence (see Supplementary Material Tab.1). Figure 5 compares the membership ratios for real and synthetic counterfactual explanations under two evaluation strategies, linear (top row) and hierarchical (bottom row), after KL divergence minimization. Under the linear selection score, the composition of synthetic ensembles closely aligns with that of the real ensembles in three out of four

Table 2 Quality scores for synthetic data across different generation methods. CTGAN outperforms the other methods, achieving the highest average score, and is thus chosen for further analysis

Dataset	CTGAN	GaussianCopula	TVAE	CopulaGAN
Credit	0.85	0.80	0.76	0.81
E-com	0.92	0.86	0.83	0.86
Iranian	0.88	0.86	0.94	0.85
Telco	0.91	0.82	0.72	0.80
Average	0.89	0.84	0.81	0.83

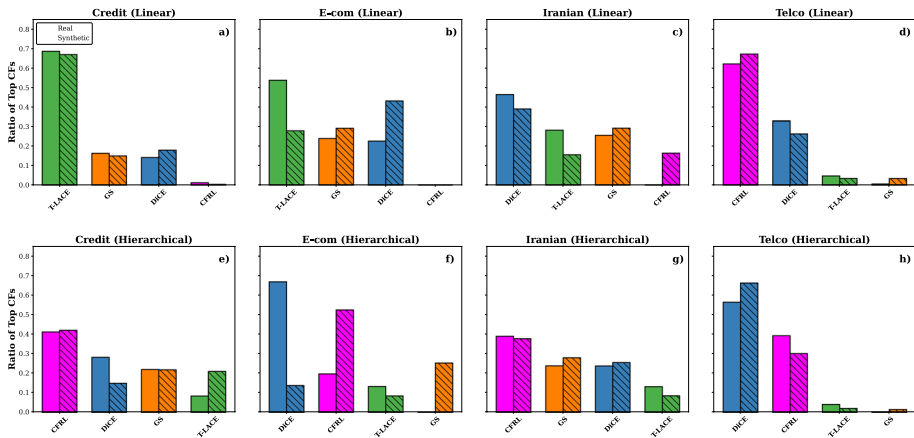


Fig. 5 Comparative analysis of membership ratios for real and synthetic datasets across evaluation methods (Linear and Hierarchical) following KL Divergence minimization. The top row illustrates membership ratios for real and synthetic counterfactuals under linear evaluation, while the bottom row represents hierarchical evaluation. Each panel corresponds to a specific dataset (Credit, E-com, Iranian, Telco), with bars indicating membership ratios for different models. Solid bars represent real counterfactuals, and hatched bars denote synthetic counterfactuals. The x-axis is sorted in descending order of membership ratio for real datasets

datasets: T-LACE dominates for Credit, DiCE for Iranian, and CFRL for Telco. The only exception is the E-com dataset, where T-LACE dominates in the real ensemble, while DiCE prevails in the synthetic one. A similar pattern emerges under the hierarchical evaluation, though the dominant method varies across datasets: CFRL leads in Credit and Iranian, while DiCE dominates in Telco. Again, E-com presents a notable discrepancy—DiCE is the most represented method in the real ensemble, whereas CFRL dominates in the synthetic counterpart. These findings suggest that the choice of selection score significantly influences counterfactual selection, enabling finer control over user-defined priorities. Furthermore, in a subset of datasets, we observe shifts in method composition between real and synthetic ensembles even when the evaluation criterion remains fixed.

4.3 Alignment between SHAP and counterfactual ensemble

The purpose of this experiment is to evaluate whether the ensemble method for generating counterfactual explanations effectively captures the key features that drive the model's decision-making process. Figure 6 presents a combined heatmap reporting Kendall's Tau correlation coefficients between the ranking of SHAP feature importances and the ranking of the most frequently changed features in the counterfactual explanations. In this heatmap, each row represents a different ensemble configuration of the linear selection score, encompassing several user-defined weight combinations as well as a configuration optimized via KL divergence minimization, while the columns correspond to the four benchmark churn datasets, with real data denoted by $\mathcal{R}$ and CTGAN-generated (synthetic) data by $\mathcal{S}$.

SHAP values, computed as the mean absolute Shapley values (Lundberg & Lee, 2017), quantify each feature's contribution to the model's output. In contrast, the ranking based on feature change frequencies reflects how often a feature is altered in the counterfactual expla-

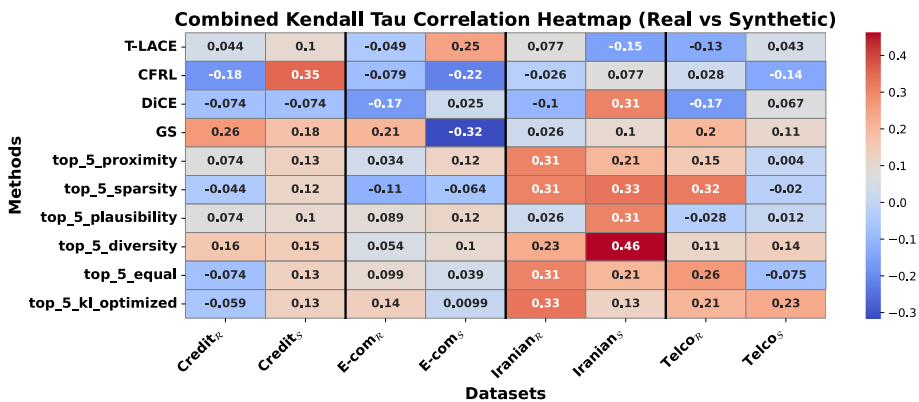


Fig. 6 Combined Kendall-Tau heatmap comparing SHAP rank correlations between individual counterfactual methods and ensemble configurations across real and synthetic datasets. Columns alternate between real (denoted as $\mathcal{R}$) and synthetic (denoted as $\mathcal{S}$) data for each dataset. Higher correlation values indicate greater alignment of ensemble-selected counterfactuals with SHAP-based feature importance rankings

nations. A higher Kendall's Tau coefficient indicates a stronger alignment between these two rankings, suggesting that the counterfactual generation method effectively identifies and prioritizes features that are critical to the model's decision-making process. The heatmap uses a coolwarm color scale, with red tones representing strong positive correlations and blue tones indicating weak or negative correlations. Importantly, our results indicate that for each dataset there is always at least one ensemble configuration capable of maintaining high correlations with the SHAP feature relevance—even when a single method is predominant, an appropriately weighted ensemble can achieve comparable performance. For example, in the Credit Churn dataset, an ensemble configuration emphasizing diversity achieves a correlation of approximately 0.16 on the real data, compared to a maximum correlation of 0.26 achieved by GS. In the synthetic Credit dataset, the configuration emphasizing diversity achieves a correlation of approximately 0.15 while individual generators such as CFRL and GS achieve a correlation of 0.35 and 0.18, respectively. Similar trends are observed across the remaining datasets.

4.4 Relationship between cluster similarity and counterfactual ensembles

We want to understand if our KL minimization methodology can help the user in devising realistic, population-specific strategies on synthetic data, which resemble the ones that would have been obtained from real data. To do so, we adopt the evaluation design that we introduced in Section 3.5. This analysis investigates the relationship between data similarity and the consistency of counterfactual behavior in both real and synthetic domains. First, we partitioned our datasets by label (churner versus non-churner) and applied K-Means clustering with $K = 5$, as determined by the Elbow Method (see Fig. 5 in the Supplementary material) independently to both real and synthetic data. Building on the findings in Sec. 4.2, we computed weight vectors that minimize the KL divergence for each combination of real and synthetic clusters and used these weights to select the ensemble with the linear selection score (the same results for the hierarchical selection score are in the Supplementary

Material Fig. 6). Subsequently, for each cluster, we computed the delta ($X' - X$) between the ensemble of counterfactuals X' and the original input data X . The Euclidean distance between the real and synthetic counterfactual deltas was then measured in order to assess whether clusters that are closer in the feature space yield more similar counterfactual behavior. In essence, our goal was to determine if structural similarity in the underlying data corresponds to greater consistency in the resulting counterfactual ensembles. Figure 7 presents a scatter plot where each point represents a pair of matched clusters. The x-axis denotes the Euclidean distance between cluster centroids, while the y-axis shows the Euclidean distance between the corresponding counterfactual deltas. Our hypothesis was that increased separation between clusters would be reflected by larger differences between the ensemble outputs, thereby confirming a correlation between the structural similarity of clusters and the consistency of their counterfactual explanations. As illustrated in Fig. 7, the data reveal a significant statistical correlation between the Euclidean distance of cluster centroids and that of the counterfactual ensembles for E-com dataset churners, Iranian dataset churners, and Telco dataset non-churners. In other words, clusters that are further apart in feature space tend to produce counterfactual ensembles that diverge more substantially.

4.5 Qualitative evaluation of explanations

Figure 8 highlights the flexibility offered by our counterfactual generation method, allowing users to steer the explanations toward different intervention strategies depending on their needs. In the top-left panel, the sparse counterfactuals demonstrate how minimal and concentrated edits on a few features can produce actionable alternatives with limited resources, ideal when straightforward and somewhat inexpensive interventions are desired. In contrast, the top-right panel shows diverse counterfactuals that explore a broader set of changes across multiple features, giving users a wider range of distinct options to consider and gain more knowledge on the user profile leading to churning. The bottom-left panel emphasizes counterfactuals that remain very close to the original instance, reflecting scenarios where subtle, incremental adjustments are preferred to minimize the effort or risk associated with

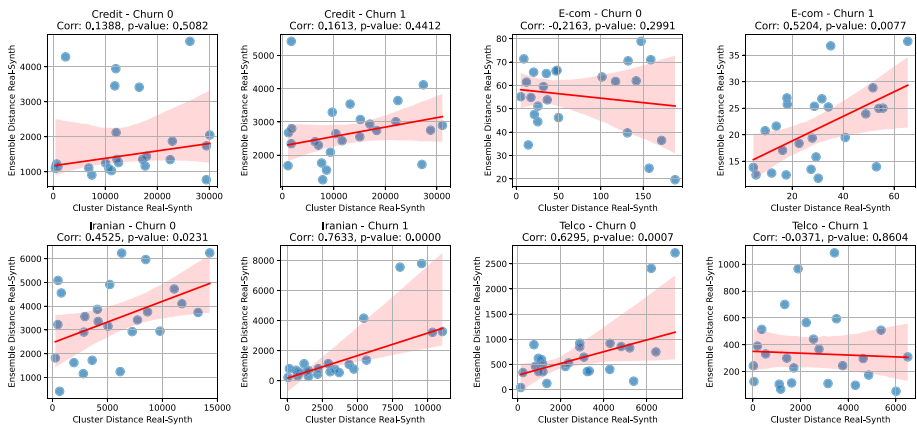


Fig. 7 Relationship between Euclidean distances of clusters of real and synthetic data and euclidean distances of ensembles counterparts

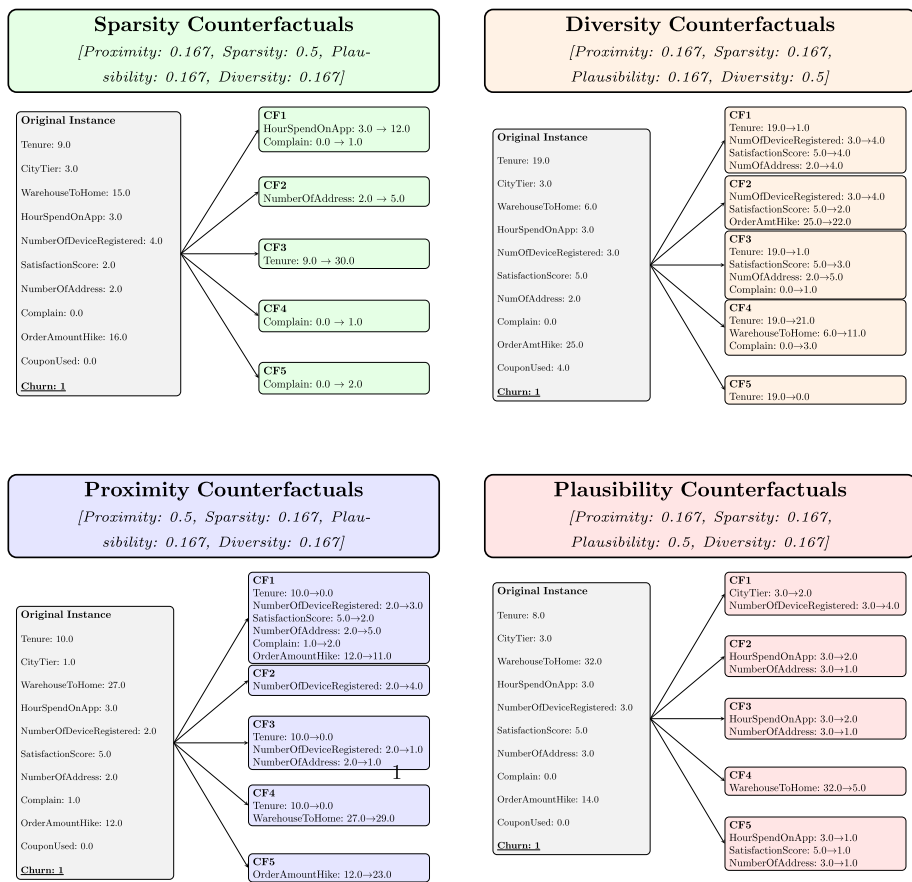


Fig. 8 Visual comparison of two counterfactual strategies from the E-commerce Churn dataset: **Top Left (green)** shows sparse explanations with minimal feature edits to reverse the churn prediction, while **Top Right (orange)** emphasizes diversity with varied and broader changes across multiple features. The sparse counterfactual **Bottom Left (blue)** prioritizes *proximity* by making small, focused changes, whereas the counterfactuals in **(Bottom Right (pink))** favors *plausibility*, offering realistic and varied alternatives across multiple features

change. Meanwhile, the bottom-right panel displays counterfactuals prioritizing realistic, coherent edits across the feature space, illustrating how the method can suggest alternatives that align with practical, plausible scenarios for churn retention strategies.

Overall, the figure demonstrates how our approach supports a wide range of intervention styles, from minimal and focused to varied and realistic, by enabling users to adjust their priorities according to their specific constraints or preferences. This flexibility ensures that stakeholders can generate explanations best suited to their operational context, empowering them with control over the balance between the metrics when designing counterfactual strategies.

5 Conclusion

In this study, we introduced a novel counterfactual ensemble selection framework that leverages multiple counterfactual generation techniques and evaluates their outputs based on a flexible and customizable scoring function. By combining these metrics through both linear and hierarchical evaluation strategies, our approach enables the selection of counterfactuals based on user preferences that aim to be more flexible and adaptable to the churn prevention task. Our method fosters interactivity with explanations and human oversight, providing different avenues of intervention. We further tailor our methodology in the context of synthetic data sharing for churn analysis, formulating an optimization problem that minimizes the KL divergence between the distributions of counterfactuals derived from real and synthetic data. This divergence minimization procedure serves as a tool to align synthetic explanations with their real-data counterparts, thus enhancing the trustworthiness of models deployed in privacy-sensitive contexts such as churn analysis. Although we adopted LightGBM as the primary predictive model in our experiments due to its strong empirical performance and relevance in real-world tabular data tasks such as churn prediction, our framework remains inherently model-agnostic. The choice of LightGBM was driven by its balance of accuracy, efficiency, and practical relevance, as confirmed by domain experts. However, the counterfactual explanation generators we employ are model-agnostic, and our approach is fully compatible with more complex or less interpretable models, such as neural networks. While our method shows strong results, there are limitations to this approach, mainly concerning the quality of the generated synthetic data. Another limitation may lie in the ability of non-technical users to navigate through the options offered by the ensemble, which require knowledge of the metrics and their impact on the output to be successfully tweaked. Further research should focus on enhancing the properties of the ensemble, incorporating new task-specific metrics tailored to the prediction challenge at hand. An additional direction for future work is to investigate whether this ensemble-based framework can be adapted from tabular data to more complex types of data, such as images. While the general idea of combining multiple counterfactual generation methods with flexible evaluation metrics remains valid, applying it to images could introduce significant challenges due to the high dimensionality and the absence of clearly defined features. Typically, generating counterfactuals for images requires advanced generative models to produce realistic and meaningful outputs, and defining clear, interpretable evaluation metrics becomes more difficult. However, by working in latent spaces and using perceptual or task-specific similarity measures, a similar ensemble approach could be developed to allow users to interactively select counterfactuals based on their preferences, making the method applicable to a wider range of data types. Finally, explainability methods are traditionally evaluated both quantitatively and qualitatively. Therefore, a rigorous user study aimed at experts in churn management would help us investigate some crucial issues in the implementation of our system: first, a user study would help us in understanding how to better communicate the explanations provided by the ensemble to the churn expert. Secondly, it would help us in investigating how a user would interact with the selection functions for the ensemble, and the effects of such changes in the interaction with the system.

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Data availability The code, datasets, and models used in this study are available on GitHub: [Counterfactual Ensembles for Interpretable Churn Prediction: From Real-World to Privacy-Preserving Synthetic Data](#).

Declarations

Conflict of interest The authors declare no competing interests.

Ethical approval and consent to participate Not Applicable.

Consent for publication Not Applicable.

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